

Quantum Field Theory in Zero Dimensions

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ABSTRACT: This article attempts to explain some of the basic features and tools of Quantum Field Theory through a baby example, a Quantum Field Theory in zero dimensions. The article is self contained and assumes no prerequisites other than high school level mathematics.

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1 Introduction

What are the fundamental constituents of the universe? One might say that it is the elements in the periodic table. But we have found a deeper model that explains (almost) everything about matter and its interactions. This is called the *Standard Model*[1]. The Standard Model is built upon a framework known as *Quantum Field Theory*(QFT). QFT is not only useful in describing the standard model. It turns out that QFT is useful in the study of Materials, Cosmology and even pure Mathematics! Unfortunately, the full machinery of QFT is rather complicated. But, we can understand some of the features of QFT using a baby example, a QFT in zero dimensions. This as we will see is just *Gaussian integrals* in a fancy wrapper. Even though calling it a QFT is an overkill, it teaches us some of the general features and techniques that we actually use in QFT.

2 What are Quantum Fields?

When we think about the constituents of matter, like the electron or a proton, the mental picture we have is that of a billiard ball. That is, particles come in lumps. One of the main ideas of QFT is that particles are not billiard balls! Instead, they

are hypothesized to be disturbances in a smooth fluid like substance that permeates the entire universe. They are called *fields*.

The simplest kind of field one can think of is called a *Scalar field*. This is an object which has a value at every point in space and time. For example, in the room you are in right now, every point has a value for temperature, and this value can change with time. So we would say that temperature is a scalar field.

Now, a Quantum Field would be a field that obeys the rules of Quantum Mechanics. “Well, What do you mean by the rules of Quantum Mechanics?”, you might ask. Let me explain. I will introduce the rules of quantum mechanics the way Feynman told us[2]. This is the story of *Path Integrals*.

2.1 Path Integrals

One standard problem of mechanics we have all done in high school is the following. Imagine you throw a ball. Given its initial position and velocity, we can calculate where the ball will land after a given amount of time. But in quantum mechanics, it is prohibited to specify the position and velocity of a particle at the same time, due to the *Heisenberg Uncertainty principle*. Therefore, if we specify the position of the particle at the initial time, we cannot specify the velocity. Hence roughly speaking, we cannot predict where the particle will end up after a certain amount of time has passed. We can only talk about the probability of the particle reaching a certain position after a certain time. A standard type of question one asks in Quantum Mechanics is the following:

If we let go of a particle at x_a at time $t = 0$, what is the probability that we find the particle at x_b at $t = T$?

This question is answered by calculating a quantity known as the *Probability amplitude*. Let us denote the probability amplitude to go from x_a at $t = 0$ to x_b at $t = T$ by,

$$\langle x_b; T | x_a; 0 \rangle \tag{2.1}$$

Once we calculate the amplitude, the probability is found by,

$$\text{Prob}((x_a, 0) \rightarrow (x_b, T)) = |\langle x_b; T | x_a; 0 \rangle|^2 \tag{2.2}$$

The main question is, how do we calculate the amplitude? This is where Feynman comes in. He told that the amplitude for the particle to take a path $y(t)$ is proportional to $e^{-S[y(t)]}$. Here $S[y(t)]$ is known as the *action* for the path. Action assigns a number to each and every path. Specifying S specifies the theory¹. Feynman’s idea is that the particle starts at x_a and then takes *every possible path* to x_b ! The

¹We will construct S for a zero dimensional QFT later on.

probability amplitude is then calculated by adding up amplitudes for every possible path,

$$\langle x_b; T | x_a; 0 \rangle = \sum_{x(t) \in \mathcal{P}} e^{-S[x(t)]} \quad (2.3)$$

Where $\mathcal{P}$ is the set of all paths from x_a to x_b .

Since we can only talk about probabilities, we can only compute averages in quantum mechanics. For example, the amplitude to take one particular path $y(t)$ is,

$$\text{Amp}(y(t)) = \frac{e^{-S[y(t)]}}{\sum_{x(t) \in \mathcal{P}} e^{-S[x(t)]}} \quad (2.4)$$

Then, the amplitude for the average value of the position at time t_1 can be calculated as,

$$\langle y(t_1) \rangle = \frac{\sum_{y(t) \in \mathcal{P}} y(t_1) e^{-S[y(t)]}}{\sum_{x(t) \in \mathcal{P}} e^{-S[x(t)]}} \quad (2.5)$$

Or to calculate the correlation between positions at different times t_1 and t_2 ,

$$\langle y(t_1) y(t_2) \rangle = \frac{\sum_{y(t) \in \mathcal{P}} y(t_1) y(t_2) e^{-S[y(t)]}}{\sum_{x(t) \in \mathcal{P}} e^{-S[x(t)]}} \quad (2.6)$$

Quantum Fields obey the same rules. The question we ask is, given a particular configuration of fields at some initial time, what is the probability amplitude to get to another configuration in a given time? The way we answer the question is also exactly the same. If we call the field ϕ , and the set of all configurations of the field to be $\mathcal{C}$, then the amplitude is given by,

$$\langle \phi_b(\mathbf{x}); T | \phi_a(\mathbf{x}); 0 \rangle = \sum_{\phi \in \mathcal{C}} e^{-S[\phi]} \quad (2.7)$$

Typically, we are interested in amplitudes in which the field configuration at $t = -\infty$ and $t = +\infty$ are given. Then the amplitude is called the *Partition function*². This is also computed the same way,

$$Z = \sum_{\phi \in \mathcal{C}} e^{-S[\phi]} \quad (2.8)$$

Where are the particles in this fancy field description? The particle is constructed out of the fields roughly as follows. First, we imagine “nothing”. Let us denote nothing or the *vacuum*, (i.e, a state with no particles) by the symbol,

$$\text{“nothing”} = |0\rangle \quad (2.9)$$

²a lot of details are swept under a huge rug!

Then, the field ϕ “creates” a particle out of nothing! This is denoted as,

$$\text{one particle} \sim \phi |0\rangle \quad (2.10)$$

Isn’t this a violation of matter and energy conservation? The answer is that the above construction is a mathematical tool that allows us to easily deal with a theory with variable number of particles. It does not mean that we actually create a particle from nothing.

3 Free Zero Dimensional QFT

As we said earlier, the fields are spread throughout the entire universe. We know that our universe is four dimensional. That is, three dimensions of space and one dimension of time. This is hard! Therefore, let us make a very drastic simplification: Let our universe be zero dimensional![3]

What would a zero dimensional universe look like? A two dimensional world is like a sheet and a one dimensional world is like a line. Continuing the analogy, a zero dimensional world is just a point! Therefore, our universe looks like this,

$$\text{Universe} = \{\text{point}\} \quad (3.1)$$

Let us now add fields to the universe. We use a real scalar field³ which has a real value at every point in the universe. Since the universe has only one point, the field is just a real number sitting at the point p . Let us denote the field by $\phi \in \mathbb{R}$.

Now, what are the possible configurations of the field? The field can take any number to be its value. Therefore, the set of all possible configurations is just the set of real numbers $\mathbb{R}$. Thus, the sum over all configurations is just an integral over real numbers.

$$\sum_{\phi \in \mathcal{C}} \equiv \int_{-\infty}^{\infty} d\phi \quad (3.2)$$

Next, we have to specify the action S . In general, action is a nice polynomial function of the fields⁴. That is, action generally looks like,

$$S(\phi) = \frac{1}{2}m^2\phi^2 + \frac{\lambda_3}{3!}\phi^3 + \frac{\lambda_4}{4!}\phi^4 + \dots \quad (3.3)$$

By “nice function” I mean that there won’t be terms like $1/\phi$. Also, m and λ are coefficients, called *couplings*. For now, we keep things simple and consider what is known as the *Free Theory*. This is the theory for which one can compute the partition function exactly. For free theory,

$$S_0(\phi) = \frac{1}{2}m^2\phi^2 \quad (3.4)$$

³since the universe has only one point, there is no notion of Lorentz transformations. Hence the only possible fields are scalars.

⁴There is no kinetic term because there is no notion of a time derivative in zero dimensions.

Let us go ahead and compute the partition function:

$$Z_0 = \int_{-\infty}^{\infty} d\phi \exp\left\{-\frac{1}{2}m^2\phi^2\right\} = \sqrt{\frac{2\pi}{m^2}} \quad (3.5)$$

How to compute this integral can be found in [4].

3.1 Feynman Diagrams

As we said earlier, computing averages is the main theme in QFT. In zero dimensional QFT, the averages are of the form,

$$\langle\phi^n\rangle = \frac{1}{Z_0} \int_{-\infty}^{\infty} d\phi \phi^n \exp\{-S_0(\phi)\} \quad (3.6)$$

There is a trick to calculate the averages quickly. The trick is due to Feynman again, and it goes by the name of *Feynman Diagrams*. First we compute,

$$Z_0(J) = \int_{-\infty}^{\infty} d\phi \exp\left\{-\left(\frac{1}{2}m^2\phi^2 + J\phi\right)\right\} \quad (3.7)$$

instead of (3.5). Here, J is called a *source*. Once we calculate (3.7), the averages are just derivatives of $Z_0(J)$. This can be easily understood. For example,

$$\begin{aligned} \frac{\partial^2}{\partial J^2} Z_0(J) &= \frac{\partial^2}{\partial J^2} \int_{-\infty}^{\infty} d\phi \exp\left\{-\left(\frac{1}{2}m^2\phi^2 + J\phi\right)\right\} \\ &= \int_{-\infty}^{\infty} d\phi \frac{\partial^2}{\partial J^2} \exp\left\{-\left(\frac{1}{2}m^2\phi^2 + J\phi\right)\right\} \\ &= \int_{-\infty}^{\infty} d\phi \phi^2 \exp\left\{-\left(\frac{1}{2}m^2\phi^2 + J\phi\right)\right\} \end{aligned} \quad (3.8)$$

Therefore, we can get $\langle\phi^2\rangle$ by taking two derivatives of $Z_0(J)$ and then setting $J = 0$ at the end! In general,

$$\langle\phi^n\rangle = \left(\frac{1}{Z_0(J)} \frac{\partial^n}{\partial J^n} Z_0(J)\right)\Big|_{J=0} \quad (3.9)$$

The integral in (3.7) can be evaluated by completing the square. We write,

$$\frac{1}{2}m^2\phi^2 + J\phi = \frac{1}{2}m^2\left(\phi + \frac{J}{m^2}\right)^2 - \frac{1}{2}\frac{J^2}{m^2} \quad (3.10)$$

Now do a substitution, $\chi = \phi + \frac{J}{m^2}$. Then (3.7) becomes,

$$Z_0(J) = \exp\left\{\frac{1}{2}\frac{J^2}{m^2}\right\} \int_{-\infty}^{\infty} d\chi \exp\left\{-\frac{1}{2}m^2\chi^2\right\} = \sqrt{\frac{2\pi}{m^2}} \exp\left\{\frac{1}{2}\frac{J^2}{m^2}\right\} \quad (3.11)$$

When we do full fledged QFT, the calculation actually proceeds similarly, but the terms involved will be rather long and will be tedious to write over and over again. Hence, Feynman introduced diagrammatic notation for the terms. They are as follows:

$$\text{---} = \frac{1}{m^2} \quad (3.12)$$

and

$$\textcircled{\text{---}} = J \quad (3.13)$$

Then overall, we can write,

$$Z_0(J) = Z_0 \exp \left\{ \frac{1}{2} \textcircled{\text{---}} \textcircled{\text{---}} \right\} \quad (3.14)$$

Where $Z_0 = \sqrt{\frac{2\pi}{m^2}}$, calculated in (3.5).

3.2 Calculation of Averages

We can now try to calculate an average. Using pictorial representation is an overkill in this simple case. But this is to give an idea of how calculations are done in QFT. Let us compute the average of four fields,

$$\langle \phi^4 \rangle = \left(\frac{1}{Z_0(J)} \frac{\partial^4}{\partial J^4} Z_0(J) \right) \Big|_{J=0} \quad (3.15)$$

We can expand $Z_0(J)$ as,

$$Z_0(J) = Z_0 \left[1 + \left(\frac{1}{2} \textcircled{\text{---}} \textcircled{\text{---}} \right) + \frac{1}{2!} \left(\frac{1}{2} \textcircled{\text{---}} \textcircled{\text{---}} \right)^2 + \dots \right] \quad (3.16)$$

We are taking four derivative of $Z_0(J)$. Since each “blob” represent a J , we only need to consider the term with four blobs. This is because we set $J = 0$ at the end. If we take any term with more than four blobs, then after taking four derivatives, there will be still blobs left. Therefore when setting $J = 0$, these terms vanish. If there are less than four blobs in the term, then taking four derivatives itself make the term vanish. Thus, we only need to calculate,

$$\frac{\partial^4}{\partial J^4} \frac{1}{2!} \left(\frac{1}{2} \textcircled{\text{---}} \textcircled{\text{---}} \right)^2 \quad (3.17)$$

But what is,

$$\frac{\partial}{\partial J} \textcircled{\text{---}} \textcircled{\text{---}} \quad (3.18)$$

Since each blob is a J and there are two ways to take a derivative on two J 's, we get,

$$\frac{\partial}{\partial J} \textcircled{\text{---}} \textcircled{\text{---}} = 2 \times \text{---} \textcircled{\text{---}} \quad (3.19)$$

The rule to remember is,

Taking a J derivative of a diagram with n blobs result in n times the same diagram, but one blob removed.

Then,

$$\begin{aligned}
\frac{\partial^4}{\partial J^4} \frac{1}{2!} \left(\frac{1}{2} \text{---} \textcircled{\cdot} \text{---} \textcircled{\cdot} \right)^2 &= \frac{\partial^3}{\partial J^3} \left(\frac{1}{2} \text{---} \textcircled{\cdot} \text{---} \textcircled{\cdot} \right) \left(\text{---} \textcircled{\cdot} \right) \\
&= \frac{\partial^2}{\partial J^2} \left[\left(\text{---} \textcircled{\cdot} \right) \left(\text{---} \textcircled{\cdot} \right) + \left(\frac{1}{2} \text{---} \textcircled{\cdot} \text{---} \textcircled{\cdot} \right) \left(\text{---} \right) \right] \\
&= \frac{\partial}{\partial J} \left[2 \left(\text{---} \textcircled{\cdot} \right) \left(\text{---} \right) + \left(\text{---} \textcircled{\cdot} \right) \left(\text{---} \right) \right] \\
&= 3 \left(\text{---} \right)^2 \\
&= 3 \times \left(\frac{1}{m^2} \right)^2
\end{aligned} \tag{3.20}$$

Thus we get,

$$\langle \phi^4 \rangle = 3 \times \left(\frac{1}{m^2} \right)^2 \tag{3.21}$$

What would happen to the expectation value $\langle \phi^{2n+1} \rangle$? From (3.16), it is clear that there are no diagrams with odd number of blobs. Hence, this average has to be zero! Also, it is not hard to verify that,

$$\langle \phi^{2n} \rangle = \frac{(2n)!}{2^n n! (m^2)^n} \tag{3.22}$$

4 Interacting Zero Dimensional QFT

Let us add interactions. One of the important examples of interactions is called the ϕ^4 interaction. This is introduced by changing the action to,

$$S(\phi) = \frac{1}{2} m^2 \phi^2 + \lambda \phi^4 \tag{4.1}$$

Roughly speaking, in full four dimensional QFT, this allows four “particles” to collide at a point. Here, we just have one point only. So this serves as a good pedagogical tool. Let us compute the full partition function, but without the source term for simplicity,

$$Z = \int_{-\infty}^{\infty} d\phi \exp \left\{ - \left(\frac{1}{2} m^2 \phi^2 + \lambda \phi^4 \right) \right\} \tag{4.2}$$

Calculating this exactly is impossible. What we do then is to compute this *perturbatively*. We assume the value of λ is small, and hence the corrections the ϕ^4 term makes to Z_0 is small. Therefore, we calculate the corrections to Z_0 order by order in λ . This is done as follows. We first expand the interaction term,

$$e^{-\lambda \phi^4} = 1 + (-\lambda \phi^4) + \frac{1}{2!} (-\lambda \phi^4)^2 + \dots = \sum_{n=0}^{\infty} \frac{(-\lambda)^n}{n!} \phi^{4n} \tag{4.3}$$

Then,

$$Z = \int_{-\infty}^{\infty} d\phi \left(\sum_{n=0}^{\infty} \frac{(-\lambda)^n}{n!} \phi^{4n} \right) \exp \left\{ - \left(\frac{1}{2} m^2 \phi^2 \right) \right\} \quad (4.4)$$

Let us exchange the sum and the integral,

$$Z = \sum_{n=0}^{\infty} \frac{(-\lambda)^n}{n!} \int_{-\infty}^{\infty} d\phi \phi^{4n} \exp \left\{ - \left(\frac{1}{2} m^2 \phi^2 \right) \right\} \quad (4.5)$$

We have already calculated this! The answer is,

$$Z = Z_0 \sum_{n=0}^{\infty} \frac{(-\lambda)^n}{n!} \frac{(4n)!}{2^{2n} (2n)! (m^2)^{2n}} \quad (4.6)$$

In this case, we could compute all the terms in the series in closed form very easily. In full QFT, it is often really hard to calculate the first order term in λ (i.e, the λ^1 term).

4.1 Asymptotic Series

In full QFT, the hope is that we compute Z (or other relevant quantities) order by order in λ and then it will be a good approximation to the full Z given by (4.2) if λ is small. In this toy example, since we have the full result to all orders in λ , we should be able to verify this. Let us use some values for the couplings. Let,

$$m^2 = 1 \quad \text{and} \quad \lambda = 0.01 \quad (4.7)$$

Then, we can calculate the partition function in (4.2) numerically to obtain the value (see appendix A),

$$Z_{\text{numerical}} = 2.44158 \quad (4.8)$$

Now, let us calculate the value of Z using the perturbation series given in (4.6). The first term is,

$$Z_{\text{perturbative}} = 2.50663 + \mathcal{O}(\lambda^1) \quad (4.9)$$

Not bad! Let us add up the first 10 terms,

$$Z_{\text{perturbative}} = 2.44291 + \mathcal{O}(\lambda^{11}) \quad (4.10)$$

The result is getting better and better! Let us be ambitious and add up the first 25 terms,

$$Z_{\text{perturbative}} = -3498.36 + \mathcal{O}(\lambda^{26}) \quad (4.11)$$

What!? This is way off from the numerical result! What happened? It turns out that the exchange of sum and integral we did in the last section is illegal! The price we have to pay is to live with this weird kind of approximation, called an *asymptotic series* approximation[3, 5]. The series gets closer and closer to the actual value as

we add up more and more terms, but only up to a certain order in λ . In this case, up to around $\mathcal{O}(\lambda^{14})$, the series gets closer and closer to the numerical value, and after that starts diverging.

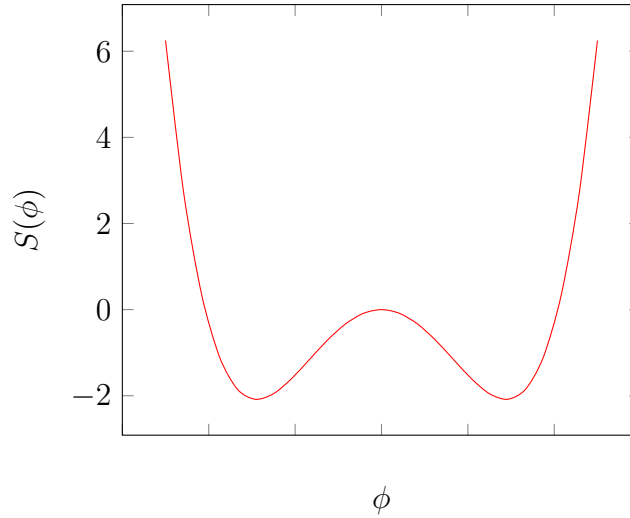
In real life, calculating even the first order diagram is hard. The best calculations we have is (I think) till order 5 or 6 in the coupling constant. Therefore, we live happily with perturbative QFT!

5 The “Higgs” Field

The field ϕ given by the action (4.1) creates a particle of mass m from the vacuum. Now we imagine a scenario where we reverse the sign in front of m^2 in (4.1). i.e, consider,

$$S(\phi) = -\frac{1}{2}m^2\phi^2 + \lambda\phi^4 \quad (5.1)$$

Well, does this mean that the mass is now $\sqrt{-m^2} = im$? This will be absurd. The issue it turns out is that we are not interpreting the “nothing” correctly. Vacuum, as we discussed before is the state with no particles. So naturally, it is the state with the lowest “energy”. That is, we first have to see what field configuration minimizes the action (5.1). This would be the actual vacuum. We can make a plot of the action to see that the minimum is clearly not at $\phi = 0$.



The actual minimum can be found by,

$$\frac{dS(\phi)}{d\phi} = 0 \quad (5.2)$$

This gives,

$$\phi_{\min} = \pm \sqrt{\frac{m^2}{4\lambda}} \quad (5.3)$$

There are two possible values for the minimum. But the vacuum can be only one. The choice is arbitrary. Let us choose the positive one as the vacuum. Then, we expand the action around this minimum,

$$\phi = \sqrt{\frac{m^2}{4\lambda}} + \chi \quad (5.4)$$

This gives (disregarding constants),

$$S = m^2 \chi^2 + 4\lambda \sqrt{\frac{m^2}{4\lambda}} \chi^3 + \lambda \chi^4 \quad (5.5)$$

Now, we see that the mass for the χ field is positive!

The interesting thing is, the action (5.1) had a symmetry of $\phi \rightarrow -\phi$. The action remained intact under this sign change. But when we re-write the action around a vacuum, there is no symmetry of $\chi \rightarrow -\chi$. Ultimately, this has to do with we arbitrarily choosing one of the two possible vacua. This is known as *spontaneous symmetry breaking*. χ roughly is the analogue of the *Higgs Boson*!

6 Summary

What we saw are baby versions of the ideas in Quantum Field Theory. We began by a bird's eye view of what quantum fields are and qualitatively understanding the path integral representation of quantum mechanics. Then we applied it to a zero dimensional QFT, which is just a Gaussian integral. We saw Feynman diagrams and their utility, and used them to compute correlation functions. We saw how perturbation theory is justified in QFT even though the series diverges ultimately. We then had a sniff at spontaneous symmetry breaking.

A Numerical Calculations

```

NIntegrate[Exp[-0.5 * x^2 - 0.01 * x^4], {x, -Infinity, Infinity}]
2.44158

Table[ $\sqrt{2 \text{ Pi}} * \frac{(-0.01)^n}{\text{Factorial}[n]} * \frac{\text{Factorial}[4 * n]}{4^n * \text{Factorial}[2 * n]}$ , {n, 0, 20}]
{2.50663, -0.0751988, 0.0131598, -0.00434273, 0.00211708, -0.00136764, 0.00110095,
-0.00106163, 0.001193, -0.00153102, 0.00220926, -0.00354085, 0.00624074,
-0.0119966, 0.0249787, -0.0560022, 0.13451, -0.344584, 0.937842, -2.70247, 8.21955}

Table[Sum[ $\sqrt{2 \text{ Pi}} * \frac{(-0.01)^n}{\text{Factorial}[n]} * \frac{\text{Factorial}[4 * n]}{4^n * \text{Factorial}[2 * n]}$ , {n, 0, m}], {m, 0, 25}]
{2.50663, 2.43143, 2.44459, 2.44025, 2.44236, 2.441, 2.4421, 2.44104, 2.44223,
2.4407, 2.44291, 2.43937, 2.44561, 2.43361, 2.45859, 2.40259, 2.5371, 2.19251,
3.13036, 0.427888, 8.64744, -17.6669, 70.7851, -240.681, 905.904, -3498.36}

```

Figure 1.

Acknowledgments

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